Meeting – June 25

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Campus Mesh Tests

- Last week, I finished setting up my campus mesh and begun testing.
- The first tests consisted of sending packets across various routes and intervals, gathering data on delivery rate and latency.
- My initial plan was to get delivery rate and latency metrics across two different routes (Goldberg to Dalplex, LSC to CHEB) at 5-second and 1-minute intervals.
- It took quite a few attempts to get this test consistently working between the Goldberg to Dalplex and, unfortunately, did not work between the LSC and CHEB.
- It could be interesting to use this data to see how Meshtastic simulations compared to real-world results, and how MeshCore performs with the same test.

Meshtastic Delivery Rate + Latency Test Results

Route: Goldberg to Dalplex

Test Type	Avg DR (%)	Avg Latency (s)	ACK Received	Bidirectional %
1 minute	51.6666667	8.187308351	28	90.32258065
5 second	26	15.02221553	53	67.94871795

- As more messages are sent throughout the mesh, delivery rate and latency decrease significantly
- Delivery rate for 1 minute is surprisingly low
- Latency is surprisingly high for both tests, could be because of the LONG_FAST setting?
- Unfortunately, for the time and effort put into these tests, they did not provide meaningful enough results. It could have been better to focus time and effort into another test.

Mesh-wide Retransmission Test

- The next test conducted was to determine how busy nodes are transmitting and retransmitting messages within a given time frame.
- Procedure:
  - Walk around campus, gathering initial local_stats telemetry data, including:
    - Uptime seconds
    - % of time since boot taken up transmitting (airUtilTx)
    - Number of packets transmitted since boot
    - Number of packets received since boot
    - Number of duplicate packets
    - Number of packets relayed for other nodes
  - Send 1 packet every 5 seconds 100 times from the Goldberg to the Dalplex.
  - Walk around campus and gather the telemetry data again.

Mesh-wide Retransmission Test Results

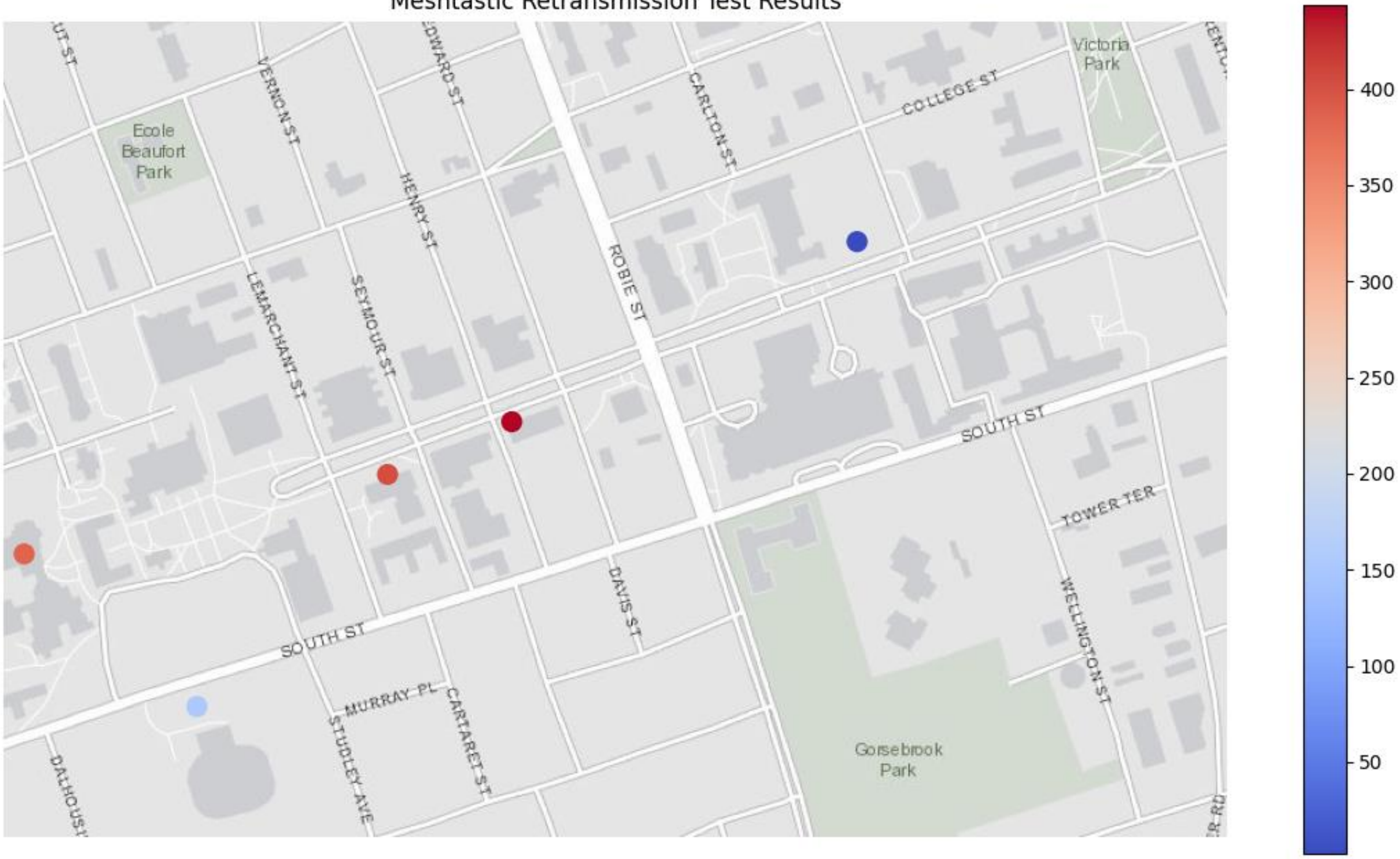
timestamp	latitude	longitude	nodeId	uptimeSeconds	channelUtilization	airUtilTx	numPacketsTx	numPacketsRx	numPacketsRxBad	numRxDupe	numTxRelay	numOnlineNodes	numTotalNodes
1782308803.6791291	44.6374710331165	-63.5875248258794	!6c73daa0	61	7.8933334	0.06255556	3	7	2	3	1	9	47
1782309202.331554	44.6369664633251	-63.5891976671482	!dadfb0cc	61	19.748333	0.12286111	6	14	1	7	5	62	62
1782310165.334353	44.6347432599171	-63.5917649968432	!dadfb8d4	61	10.715	0.05611111	2	5	3	1	1	23	66
1782310732.606525	44.6362052985069	-63.5940921770574	!dadfb008	122	7.621667	0.06408334	2	8	4	2	1	2	79
1782312597.389235	44.6391960574960	-63.5828730326646	!dadfb03c	61	N/A	0.028055554	1	N/A	N/A	N/A	N/A	41	41
1782315554.031728	44.6374710331165	-63.5875248258794	!6c73daa0	6844	2.0033333	5.9859166	447	665	156	400	64	18	51
1782316035.450979	44.6369664633251	-63.5891976671482	!dadfb0cc	6902	4.14	5.2986665	409	1036	132	406	399	70	70
1782316761.518246	44.6347432599171	-63.5917649968432	!dadfb8d4	6662	2.0033333	1.7945001	155	812	70	267	133	24	66
1782317435.696922	44.6362052985069	-63.5940921770574	!dadfb008	6784	7.5516663	4.1583333	386	832	95	261	385	24	79
1782318815.515288	44.6391960574960	-63.5828730326646	!dadfb03c	6242	N/A	0.028833333	3	N/A	N/A	N/A	N/A	41	41
	44.6374710331165	-63.5875248258794	!6c73daa0	6783		5.92336104	444	658	154	397	63	9	4
	44.6369664633251	-63.5891976671482	!dadfb0cc	6841		5.17580539	403	1022	131	399	394	8	8
	44.6347432599171	-63.5917649968432	!dadfb8d4	6601		1.73838899	153	807	67	266	132	1	0
	44.6362052985069	-63.5940921770574	!dadfb008	6662		4.09424996	384	824	91	259	384	22	0
	44.6391960574960	-63.5828730326646	!dadfb03c	6181		0.000777779	2	N/A	N/A	N/A	N/A	0	0

Note: !dadfb03c (CHEB) did not have the default telemetry settings turned on

Mesh-wide Retransmission Test Results

latitude	longitude	nodeId	uptimeSeconds	airUtilTx	numPacketsTx	numPacketsRx	numPacketsRxBad	numRxDupe	numTxRelay	numOnlineNodes	numTotalNodes
44.6374710331165	-63.5875248258794	!6c73daa0	6783	5.92336104	444	658	154	397	63	9	4
44.6369664633251	-63.5891976671482	!dadfb0cc	6841	5.17580539	403	1022	131	399	394	8	8
44.6347432599171	-63.5917649968432	!dadfb8d4	6601	1.73838899	153	807	67	266	132	1	0
44.6362052985069	-63.5940921770574	!dadfb008	6662	4.09424996	384	824	91	259	384	22	0
44.6391960574960	-63.5828730326646	!dadfb03c	6181	0.000777779	2	N/A	N/A	N/A	N/A	0	0

Meshtastic Retransmission Test Results



Mesh-wide Retransmission Test

- There were some issues with the test:
  - There is supposed to be another node (!dadfb6c8) in the McCain Arts Building Room 2130 but unfortunately, the doors to the classroom have been locked. Fortunately, the node is still active and is still part of the mesh.
  - While I thought that the node in the Collaborative Health Education Building was re-flashed recently and had all the default settings, this was not the case, and I only realized this until after the test was completed. Because the telemetry settings were turned off, I was only able to see the airUtilTx and number of packets sent, which were low because the default background settings were turned off.
- This test could be re-done in the future with these changes for more data and accurate results.

Next Week

- Battery Consumption Testing
  - Observing how much battery is consumed on both nodes based on default conditions (overnight)
- Mobile Routing Test
  - Observing how Meshtastic reacts to moving nodes, how routes dynamically update when a path is broken and how this affects other statistics like delivery rate and latency.
- MeshCore
  - Doing the previous tests (and other tests) on MeshCore firmware to compare how changes in architecture impact results in an urban environment.